Medical image denoising

Checkpoint 1 Background and model selection

CPSC 4430 and 6430 | Fall 2026 | T3

Plan noisy-only MRI denoising that reproduces NGPS target sampling and tests an improvement. Present at least one paper and begin a download or ML-method introduction. Checkpoints 2 and 3 establish baselines and evaluate structure and runtime.

Starting point: Use public BrainWeb T1 volumes or an instructor-released volumetric alternative; read the T3 overview and guide. Clean references may generate noise and score frozen results, never train, guide or select models. Synthetic phantom results do not establish clinical performance.

2026 paper links: NGPS (ECCV); Learning from Physical Noise (ECCV); Cryo EM Score Matching (ECCV); SPIDER (Nature Communications). Full titles and local PDFs are listed in Paper_to_Checkpoint_Map.txt.

Required work

1 Present at least one paper

Choose at least one of the four linked papers for your group to present. Explain its method pipeline and major contributions. Explicitly identify its datasets and access routes or simulator/data-generation procedure; ML models, inputs, outputs, supervision and loss; and evaluation plan, including tasks, splits or held-out units, baselines, metrics and compute budget. Identify unstated details and distinguish the paper's protocol from your proposed course evaluation. Explain the noisy-pair, neighboring-slice, structural-prior or acquisition-redundancy assumptions required by the selected method.

2 Begin the practical work

Complete the following practical start:

1. begin downloading data and show URLs, files or progress logs;
2. begin simulator setup and show versions, commands and setup/error logs;
3. give a brief introduction to your chosen ML method with inputs, outputs, loss, code entry points and an initial training plan.

A completed download, working simulator pilot or trained model is not required at Checkpoint 1. Suitable evidence is BrainWeb download progress or a NGPS method introduction.

3 Trace the NGPS supervision pipeline

Map guide.py, sampler.py, losses.py and nafnet.py to filtered-guide generation, static/changing masks, local patch matching, Top K raw-neighbor target retrieval and hybrid/regional losses. Guides choose correspondences; raw noisy neighbors supply targets. Identify anatomical misalignment and noise correlation as possible failures.

4 Choose data and matched baselines

Plan at least 10 subjects/phantoms, roughly 6/2/2 by subject; keep slices, patches and noise draws together. Use Gaussian sigma 0.03, 0.06 and 0.10 on [0,1] intensities. Compare NLM, masked-pixel blind-spot, unmasked same-coordinate adjacent targets and NGPS; share NAFNet across deep objectives, or document a U-Net port.

5 Predeclare selection and evaluation

Use noisy-only validation or a fixed schedule. Plan subject-level PSNR/SSIM with data_range=1, HFEN/edge metrics and complete inference timing. Select one NGPS improvement with

matching/component ablations. Define the under-10-minute GPU workload, compute, roles and timeline.

Submission and assessment

Submit one group package on Canvas by class time and give an 8-minute group presentation. Use the items below as the submission checklist.

- A 2-page report with the selected paper's datasets, ML supervision and evaluation protocol, public MRI sources, method/code map, subject splits, noise/metrics, improvement, practical-start status, roles and timeline.
- Slides for an 8-minute presentation with the paper's pipeline and contributions, datasets, ML method, evaluation plan, noisy-only training diagram and practical-start evidence.
- A start-evidence package: download/progress log, setup log if relevant, or brief ML-method introduction with inputs, outputs, losses, code references and training plan. Include a draft subject manifest and reference-separation plan; pilots are optional.

Completion criteria

The group presents at least one paper's dataset, ML methods and evaluation plan, submits evidence for at least one practical start, and defines a feasible reproduction and held-out course evaluation. Setup may remain in progress; record the next action and owner. Clean references are excluded from training, guide construction and selection.

Grading rubric

Criterion	Points	Evidence
Understanding of the problem	30	Correct MRI scope, noise assumptions, neighboring-slice geometry and reference separation.
Quality of literature summary	25	At least one paper presented with datasets, ML supervision, evaluation protocol and contributions.
Soundness of model choice	30	Feasible NGPS reproduction, matched baselines, independent evaluation and one evidenced practical start.
Clarity and teamwork plan	15	Clear training-input diagram, responsibilities and three-checkpoint plan.

Total: 100 points.

Team responsibilities

Groups may have up to 4 students. Each report must list who did what. Submit a short peer evaluation after each checkpoint; individual scores may be adjusted by up to 10% in either direction. CPSC 6430 students lead the model-comparison and validation-plan sections.

Next step

Checkpoint 2 builds the noisy subject-split dataset and reproduces NLM, blind-spot and naive-neighbor baselines. Checkpoint 3 reproduces NGPS sampling, tests one improvement and evaluates anatomy and runtime.

Use released development data for learning and tuning. Instructor-held benchmark data and labels remain separate. Optional extensions should follow completion of the required work.